UNRAMIFIEDNESS AT A POINT WITH A SPECIALIZING DOMINANT POINT OVER A UNIBRANCH BASE IMPLIES ETALENESS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10465-10477 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10479-10493, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Unramified morphisms, geometrically unibranch schemes, specializations, and the flatness criterion for etaleness; the argument is retained.

Source reference. Original lemma 0GSC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10493–10493. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $x \in X$ with image $y \in Y$. Assume*

(1) *Y is integral and geometrically unibranch at y,*
(2) *f is locally of finite type,*
(3) *there is a specialization $x' \rightsquigarrow x$ such that $f(x')$ is the generic point of Y,*
(4) *f is unramified at x.*

Then f is étale at x.

Proof. We may replace X and Y by suitable affine open neighbourhoods of x and y. Then Y is the spectrum of a domain A and X is the spectrum of a finite type A-algebra B. Let $\mathfrak{q} \subset B$ be the prime ideal corresponding to x and $\mathfrak{p} \subset A$ the prime ideal corresponding to y. The local ring $A_\mathfrak{p} = \mathcal{O}_{Y,y}$ is geometrically unibranch. The ring map $A \to B$ is unramified at $\mathfrak{q}$. Also, the point x' in (3) corresponds to a prime ideal $\mathfrak{q}' \subset \mathfrak{q}$ such that $A \cap \mathfrak{q}' = (0)$. It follows that $A_\mathfrak{p} \to B_\mathfrak{q}$ is injective. We conclude by More on Algebra, Lemma 0GSC. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.